Binomial Distribution

1. $b(x; n, p) = c_x^n p^x q^{n-x}$, $x = 0, 1, 2, \ldots, n$

Hypergeometric Distribution

2. $P(X = x) = h(x; N, n, k) = \frac{(C_x^k)(C_{n-x}^{N-k})}{C_n^N}$, $\max\{0, n-(N-k)\} \leq x \leq \min\{n, k\}$

Poisson Distribution

3. $P(x; \lambda t) = \frac{(\lambda t)^x e^{-\lambda t}}{x!}$, $x = 0, 1, 2, \ldots$

4. Geometric Distribution

$g(x; p) = p\, q^{x-1}$, $x = 1, 2, 3, \cdots$

5.Negative Binomial Distribution

$b^*(x; k, p) = {}_{x-1}C_{k-1}\ p^k q^{x-k}$, $x = k, k+1, k+2, \ldots$

Multinomial Distribution

6. $f(x_1, x_2, \ldots x_k; p_1, p_2, \ldots, p_k, n) = \frac{n!}{x_1! \times x_2! \ldots \times x_k!} \times p_1^{x_1} \times p_2^{x_2} \times \ldots \times p_k^{x_k}$

Multivariate Hypergeometric Distribution

7. $f(x_1, x_2, \ldots, x_k; a_1, a_2, \ldots, a_k, N, n) = \frac{(C_{x_1}^{a_1})(C_{x_2}^{a_2})\ldots(C_{x_n}^{a_n})}{C_n^N}$

Law of Total Probability

8. $P(B) = \sum_{i=1}^{n} (A_i \cap B) = \sum_{i=1}^{n} P(A_i)P(B|A_i)$

9.Bayes' Theorem

$P(A_i|B) = \frac{P(A_i)P(B|A_i)}{\sum_{i=1}^{n} P(A_i)P(B|A_i)}$

Permutations

10. ${}_nP_r = \frac{n!}{(n-r)!}$

Multinomial Coefficient

11. $\frac{n!}{n_1!n_2! \cdots nk!}$ Or $\binom{n}{n_1, n_2, \ldots, n_r} = \frac{n!}{n_1!n_2! \cdots n_r!}$

Combinations

12. ${}_nC_r = \frac{n!}{r!(n-r)!}$

Circular permutation

13. $(n-1)!$